Hyperspectral imaging using deep learning in wheat diseases: (Review)

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***Abstract*—Wheat is a significant staple crop worldwide, but its cultivation is challenged by various factors such as pests, diseases, climate change, lack of water, and yield enhancement. To prevent wheat infections, nondestructive and self-sufficient remote detection methods are necessary, which are sensitive and reliable. Hyperspectral imaging with deep learning is a powerful method for identifying diseases early, mapping them accurately, and evaluating their severity in plants in an unbiased way. This method involves data collection, preprocessing, and analysis of Hyperspectral images to identify wheat diseases. The article also presents several Hyperspectral imaging datasets and relevant research on using this technology for detecting wheat illnesses. The integration of spectroscopy and imaging in Hyperspectral imaging represents a novel approach to achieving an in-depth and comprehensive understanding of plant life. Moreover, deep learning, a subset of artificial intelligence, has gained considerable attention due to its ability to process vast and intricate data sets. By synergistically merging these two advanced technologies, researchers and scientists can significantly enhance the analysis of hyperspectral imagery, thereby enabling improved crop monitoring, disease identification, and yield projection in wheat farming This article summarizes the use of deep learning techniques in Hyperspectral imaging for wheat crops. Hyperspectral imaging is a non-destructive way to study crops but generates a lot of data that can be difficult to analyze. Deep learning automates the analysis process and improves accuracy. However, it requires large and diverse datasets, complex model development, and computational resources. This report provides information on the benefits and challenges of using deep learning in Hyperspectral imaging for wheat crops.**

***Index Terms*—Hyperspectral imaging, agriculture disease, wheat disease, deep learning**

I. Introduction

Wheat is a vital staple crop globally, playing a critical role in preserving food security and contributing to trade and economics. It faces numerous challenges such as pests, diseases, climate change, water scarcity, and the need for yield improvement [1]. To combat wheat infections, researchers are focusing on developing disease-resistant cultivars [2]. However, traditional visual identification of plant diseases is laborious and time-consuming, hindering disease management and fungicide efficiency [3]. To address these difficulties, non-destructive and autonomous remote detection technologies with high sensitivity and reliability are required [4]. Hyperspectral imaging is a technique that captures spectral information across a broad range of wavelengths, allowing insights into plant health, stress, and nutrient levels with high precision.[3] Recent advancements in deep learning, such as convolutional neural networks (CNNs) and recurrent neural networks (RNNs), have proven effective in extracting meaningful features from hyperspectral data, leading to more accurate classification and prediction models.[4] Modern sensing techniques, such as hyperspectral imaging and deep learning models, offer valuable tools for early disease detection, disease mapping, and objective assessment of disease severity. These techniques hold great potential for improving disease management strategies, optimizing resource allocation, and minimizing crop losses due to plant diseases.[5] The report provides a comprehensive explanation of hyperspectral algorithms, data processing analysis, and articles on wheat diseases utilizing hyperspectral imaging.[3]

II. Benefits of Implementing Deep Learning in Hyperspectral Imaging for Wheat Agriculture

A. *Improved Crop Monitoring*

Crop health may be continuously monitored thanks to the real-time analysis of hyperspectral pictures using deep learning systems. This can assist farmers in identifying early indicators of illness or stress so that prompt remedial action can be taken.

B. *Enhanced Disease Detection*

Plant disease early detection can be facilitated by deep learning algorithms that can recognize minute differences in hyperspectral pictures. Farmers may reduce crop losses and slow the spread of illnesses by taking this proactive strategy.

C. *Accurate Yield Prediction*

Deep-learning algorithms are capable of more accurate crop production prediction by evaluating hyperspectral data throughout the growing season. Farmers may use this information to plan their harvests, store their products, and develop marketing campaigns with more knowledge.

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D. *Efficient Nutrient Management*

By analyzing hyperspectral data, deep learning algorithms may be used to assess the nutritional content of wheat crops. With the use of this data, farmers can minimize their environmental effects, cut expenses, and apply fertilizer more efficiently.

III. Difficulties in Applying Deep Learning for Wheat Crop Hyperspectral Imaging

- **Data Collection and Preprocessing:** Gathering and preparing massive hyperspectral datasets can take a lot of effort and time. Accurate deep learning model training requires a variety of labeled training data sets.
- **Model Complexity and Training:** The training of deep learning models frequently necessitates a large investment of time and computing power. Refinement of these models for particular wheat crop uses requires significant computer power and skill.
- **Explainability and Interpretability:** Deep learning models frequently function as "black boxes."

IV. Algorithms and Deep Learning Techniques Used to Hyperspectral Images for Agricultural Purposes

In recent years, the application of deep learning techniques in hyperspectral imaging has shown great potential for enhancing agricultural practices, particularly in wheat crops. Hyperspectral imaging refers to the process of capturing and analyzing a wide range of wavelengths in the electromagnetic spectrum, enabling the identification of unique spectral signatures associated with various objects and materials. This advanced imaging technique coupled with deep learning algorithms has revolutionized the way we approach crop monitoring and management. Various deep-learning algorithms have been successfully used in hyperspectral imaging for agricultural purposes. We have mentioned the most famous ones

A. *Convolutional Neural Networks (CNNs)*

CNNs are artificial neural networks that learn and extract relevant features from input images. They consist of convolutional and pooling layers that work together to classify images based on their visual content. CNNs have been successfully applied in various domains, including computer vision and speech recognition. By training a CNN model on hyperspectral images, it can classify different attributes of wheat crops, such as disease detection, nutrient deficiency, or weed identification. CNNs capture spatial dependencies within spectral data[6].

B. *Recurrent Neural Networks (RNNs)*

Recurrent Neural Networks (RNNs) can process sequential data by capturing relationships between elements in a sequence. Unlike feedforward neural networks, RNNs have feedback connections that allow information to persist and be shared across time steps, enabling them to model temporal dynamics and learn from sequential data. RNNs are commonly

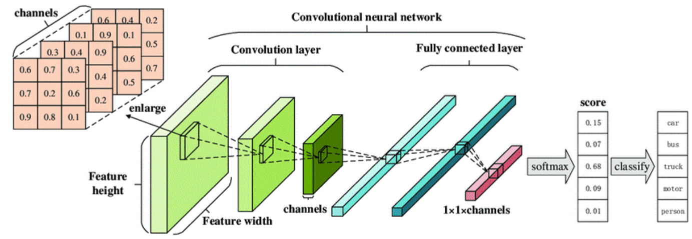

Fig. 1. Architecture of a Convolutional Neural Network (CNN)[7].

used in natural language processing, machine translation, speech recognition, and sentiment analysis.

RNNs are used in hyperspectral imaging to predict crop behaviour, such as growth stages or yield estimates. By processing hyperspectral images as a sequence of spectral bands over time, RNNs can learn patterns. This helps farmers make informed decisions about irrigation, fertilization, and harvesting[8].

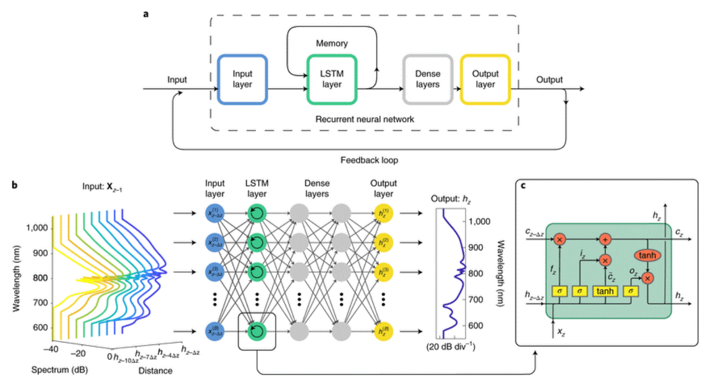

Fig. 2. Recurrent neural networks a, Schematic of the RNN architecture used, showing the input layer, the LSTM recurrent layer, two hidden (dense) layers, and the output layer[9].

C. *Generative Adversarial Networks (GANs)*

GANs are deep learning models consisting of two components: a generator network and a discriminator network. GANs are trained adversarial, where the generator and discriminator networks compete to improve both models. They have found success in image synthesis, style transfer, and data augmentation. In hyperspectral imaging, GANs enable the generation of synthetic images with specific characteristics, reducing the need for extensive labelling and allowing for accurate deep-learning models[10].

D. *Transfer Learning*

Transfer learning is a machine learning technique that applies the knowledge gained from training one model to another related task or domain. It involves using pre-trained models to improve performance on a smaller target dataset. Transfer learning is commonly used where the target dataset is small or lacks sufficient annotations. It has been successfully applied in various domains, including agriculture, where it

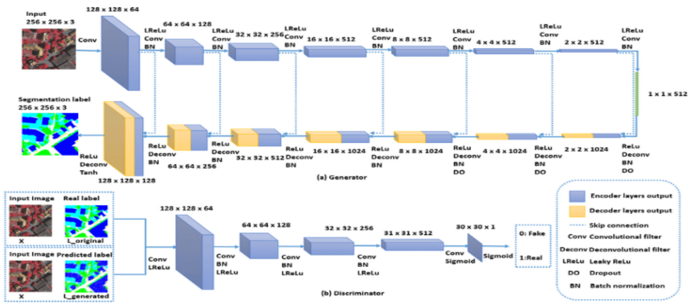

Fig. 3. The architecture of the first GAN network: (a) the generator and (b) the discriminator[11].

can significantly reduce the computational requirements and training time for hyperspectral imaging tasks[12].

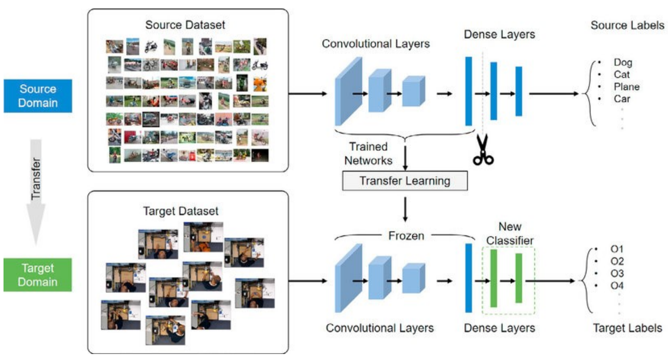

Fig. 4. The architecture of our transfer-learning model[13].

Considering the specific objective of exploring different deep-learning approaches and algorithms for hyperspectral imaging in wheat crops, a combination of CNNs and RNNs seems most suitable

V. Deep Learning in Hyperspectral Imaging Analysis and Classification

Hyperspectral imaging is a method for gathering and analyzing spectral data from a scene or item over a broad spectrum of electromagnetic frequencies. This method has the potential to be a potent instrument for acquiring thorough data on the well-being, development, and general quality of wheat crops. The analysis and classification of hyperspectral pictures may be improved even further by utilizing deep learning strategies, which employ artificial neural networks to interpret complicated patterns and extract significant characteristics from data. Hyperspectral imaging of wheat crops can be improved by utilizing deep learning in image analysis and classification. One major advantage of using deep learning algorithms is their ability to efficiently process large amounts of hyperspectral data, which leads to more effective analysis of crop conditions. These algorithms also automatically identify relevant features in the hyperspectral images such as chlorophyll concentration, leaf area index, and disease symptoms, without the need for manual identification. Furthermore, deep learning algorithms can classify wheat crops into different categories based on their health status, growth stage, or specific attributes. This classification can help farmers make informed decisions regarding disease management, fertilizer application, and irrigation strategies. It can also aid researchers in studying the impact of various factors on wheat crop development. To implement deep learning in image analysis and classification for hyperspectral imaging of wheat crops, a suitable approach would be to build a convolutional neural network (CNN) architecture. CNNs are well suited for image data analysis due to their ability to learn spatial hierarchies of features. The network can be trained using a large dataset of labelled hyperspectral images, with each image having corresponding ground truth information about the wheat crop's characteristics. During training, the CNN would learn to extract relevant features from the hyperspectral images and map them to the corresponding crop attributes. This process involves iterative optimization, where the network adjusts its internal parameters to minimize the prediction errors. After training, the CNN can be utilized to analyze and classify new, unseen hyperspectral images of wheat crops accurately[14] [15] [16].

A. *Hyperspectral imaging in wheat using deep learning*

Hyperspectral imaging in agriculture significantly extends the range of farming issues and applications that can be addressed using remote sensing. Almost every farming issue (weeds, diseases, nutrient deficiency, etc.) changes the plant's physiology, affecting its reflective properties. Hyperspectral imaging is a non-invasive technique that shows promise in early diagnosis and pathogenesis monitoring of wheat diseases. This allows for precise crop protection. The use of hyperspectral imaging and various illnesses affecting wheat crops will now be discussed in several publications.

B. *state of the art in hyperspectral imaging using deep learning*

In recent years, there has been significant progress in the field of hyperspectral imaging (HSI) with the application of deep learning techniques. The use of deep learning has revolutionized computational spectral imaging methods, allowing for advanced analysis and classification of hyperspectral data. One notable article by Deepak Kumar and al. [20] provides a comprehensive overview of the advances in hyperspectral image and signal processing. The authors review the state of the art in HSI and highlight the latest developments in the field. Their work covers various aspects of HSI, including classification and processing techniques. Another important study by Li and al. [18] focuses specifically on deep learning for HSI classification. The researchers thoroughly analyze the strengths and weaknesses of the most prominent deep learning methods used in HSI classification. Their comprehensive review provides valuable insights into the current state of the art in this area. Furthermore, a paper by Zhang and al. [19] explores deep learning-based methods for HSI classification with a focus on few-shot learning. The authors aim to provide a comprehensive review of state-of-the-art deep learning techniques in the context of HSI classification. Their research sheds light on the latest advancements and their potential applications. In addition, an article by the Nature Journal [17] reviews state-of-the-art deep learning-empowered computational spectral imaging methods. This comprehensive review covers various aspects of deep learning in HSI and provides an in-depth understanding of the latest techniques and their applications. Finally, a survey conducted by Alberto Signoroni and al. [21] aims to give an overview of the application of deep learning in the context of hyperspectral data processing. The survey provides insights into the current state of the art in the field and describes the potential benefits and challenges of using deep learning in HSI. Overall, these articles and surveys contribute to the current state of the art in hyperspectral imaging using deep learning.

VI. Relevant articles that focus on hyperspectral imaging in wheat

In recent years, hyperspectral remote sensing has emerged as a powerful tool for capturing detailed information about various aspects of wheat plants. One area of focus has been the assessment of nitrogen (N) status in wheat plants in real-time and non-destructively [22] [23]. Researchers have found that hyperspectral imaging can effectively capture information related to the nitrogen content in wheat plants, providing valuable insights for optimizing fertilization strategies and improving crop productivity.

Another study utilized hyperspectral imaging to examine the impact of Fusarium head blight (FHB) on wheat kernels [24]. By evaluating the damage caused by FHB, researchers were able to assess the severity of the disease and potentially develop strategies for its management. Hyperspectral imaging proved to be a valuable tool for this purpose.

Furthermore, the potential of hyperspectral analysis in wheat has been explored in terms of its non-destructive capabilities [24]. By analyzing individual wheat plants using field hyperspectral imaging, researchers were able to assess various parameters related to plant health and quality without causing any damage. This approach holds promise for monitoring crop conditions and making timely interventions to maximize yield. In summary, several articles have investigated the application of hyperspectral imaging in wheat. These studies have focused on evaluating nitrogen status in wheat plants, assessing the damage caused by FHB, and exploring the non-destructive potential of hyperspectral analysis. Through the use of deep learning techniques, hyperspectral imaging has demonstrated its ability to provide valuable insights for improving crop management strategies and optimizing wheat production. We have been searching about the uses of hyperspectral imaging using deep learning in wheat crops, and we have found several articles that talk about the diseases that infect the wheat crops we mentioned some of them.

The first article [25] applies a deep neural network classification algorithm to hyperspectral images to accurately identify disease areas. By analyzing the spectra of the pixels, the algorithm can effectively discern the presence of diseases. The study focuses on the classification of diseases in crops through the use of hyperspectral imaging and deep learning techniques. The study suggests that the developed classification algorithm can be used for future research on Fusarium head blight disease prediction and larger-scale disease assessment based on airborne hyperspectral remote sensing.

The second article [26] highlights the promising use of convolutional neural networks (CNNs) as tools for red-green-blue (RGB) image or spectral cereal classification. CNNs have shown potential in accurately classifying and analyzing cereal crops based on RGB images or spectral data.

The third article [23] discusses the application of hyperspectral remote sensing in capturing real-time and non-destructive information about the nitrogen (N) status in wheat plants. By utilizing hyperspectral imaging, valuable insights about the N status of wheat crops can be obtained.

The fourth article [27] explores the utilization of deep learning models in automatically extracting features related to Fusarium head blight (FHB) in wheat plants. Through the generation of images from single lines, deep learning techniques can effectively extract FHB features and aid in the detection of this harmful disease in wheat crops.

The article [28] discusses the use of hyperspectral crop reflectance data to detect and estimate fungal disease severity in wheat. The study proposes a method that involves pre-normalization and classification steps. In the pre-normalization step, the hyperspectral data is normalized to reveal and magnify the spectral characteristics caused by changes in plant condition due to the disease. Field measurements are used to build a reference dataset, and the same pre-normalization procedure is applied to new unclassified hyperspectral data. In the classification step, a nearest-neighbor classifier is used to discriminate between healthy and diseased plants and quantify the damage levels for the diseased ones. The study shows a high correlation between the classification results and field assessments of disease severity, indicating the effectiveness of this approach. The article also discusses the spectral characteristics observed in the crop and disease-specific signatures, such as a flattening of the green reflectance peak and a decrease in reflectance in the NIR region. Overall, the method proves to be useful and efficient for estimating fungal disease severity in wheat using hyperspectral crop reflectance data.

The article [29] discusses the use of hyperspectral imaging technology to detect wheat powdery mildew and assess disease severities in crops for effective pesticide application. The study focuses on differentiating background factors, such as wheat ears and shadowed leaves, to accurately identify infected and healthy plant leaves. By comparing spectral differences between these factors, the researchers identified 23 sensitive bands and calculated five vegetation indices and three red edge parameters. These features were used to develop a prediction model using the classification and regression tree (CRT) method. The accuracy of the model was assessed through cross-validation, with shadowed leaves being perfectly recognized, and healthy and infected leaves and wheat ears being identified with high rates of accuracy. However, detecting mild infection remains a challenge. The study highlights the importance of early detection and data collection in the early-infected stage to effectively guide spray control of wheat powdery mildew.

The article [30] discusses the use of hyperspectral and chlorophyll fluorescence imaging for the early detection of plant diseases, with a focus on Fusarium infections in wheat. The study aims to improve the detection of Fusarium head blight (FHB) in cereal production, particularly in wheat. The symptoms of FHB can be detected using image analysis, which allows for mapping the occurrence and extent of Fusarium infections. The article explores the characteristics, requirements, and limitations of using hyperspectral and chlorophyll fluorescence imaging for Fusarium detection in both field and laboratory settings. The modification of spectral signatures due to fungal infection allows for detection using hyperspectral imaging, while the decreased physiological activity of tissues resulting from Fusarium impacts provides the basis for chlorophyll fluorescence imaging analyses. The article also discusses the advantages and disadvantages of these imaging techniques for head blight detection. Overall, the study highlights the potential of hyperspectral and chlorophyll fluorescence imaging for the early detection of Fusarium infections in wheat.

This article [31] explores the use of hyperspectral imaging (HSI) to detect varying degrees of damage in wheat kernels caused by Fusarium head blight (FHB). The study collected wheat kernel samples with different levels of Fusarium infection and acquired hyperspectral images of the kernels. The spectral data obtained from the images were pre-processed and optimal wavelengths were selected. The study found that the spectral data alone had the potential to identify the degree of damage by Fusarium, but the performance of the models was not satisfactory. Therefore, image features were extracted based on the optimal wavelengths and fused with the spectral features. The fusion of spectral and image features improved the accuracy of the analysis for identifying the degree of damage. The study concludes that the integration of spectroscopy and image analysis using HSI can provide a fast and accurate method for detecting Fusarium damage in wheat kernels.

Hyperspectral imaging was used in the article [32] to identify wheat yellow rust, a common crop disease. Researchers used a threshold to extract wheat leaf pixels, effectively eliminating background interference. The study identified suitable wavebands for disease identification and effective spectral indices, such as NDVI and GNDVI, for distinguishing between healthy and diseased leaves. The study highlights the potential of hyperspectral imaging for early detection and monitoring of the disease.

The article [33] discusses the use of hyperspectral imaging and deep learning for the early detection of plant viral diseases, with a specific focus on grapevine vein-clearing virus (GVCV). The authors highlight the importance of early identification of plant diseases in agriculture due to climate change and its impact on virus breeding and transmission. They propose a method that combines hyperspectral imaging, spectral statistics, and machine learning algorithms to accurately identify GVCV-infected vines. The study utilizes hyperspectral data cubes with a limited number of samples and applies 2D-CNN and 3D-CNN architectures as feature extractors. Traditional machine learning algorithms such as support vector machine (SVM) and random forest (RF) classifiers are used for training and classification tasks. The authors address the challenge of small training set sizes for hyperspectral image-wise models by leveraging the joint spatial-spectral correlation information. Figure 5 mentions the architecture used in this study

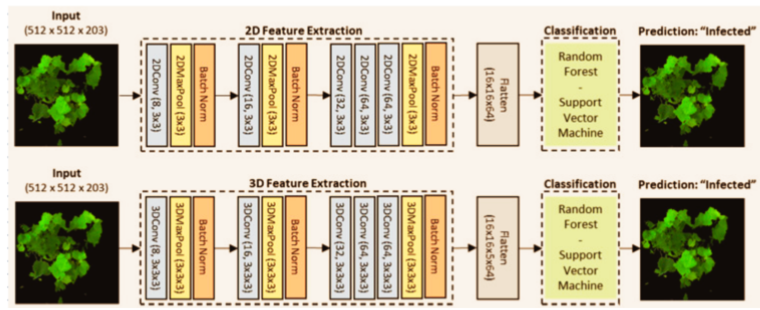

Fig. 5. The architecture diagram for the image-wise classification.[33].

The results show that the spectral signals of GVCV vines were more discriminative in the visible (VIS) wavelength range, with changing leaf pigment constituents playing a role. The study identifies specific indices, such as FRI1, PSRI, WSCT, and AntGitelson, as critical features for classifying healthy and GVCV-infected vines. The SVM classifier achieved high accuracy in the training process, ranging from 82.13% to 96.75% in 5-fold cross-validation.

Overall, the study demonstrates the potential of hyperspectral imaging and deep learning techniques for early detection and classification of plant viral diseases, specifically focusing on GVCV in grapevines.

The article [34] discusses the challenges and limitations of using Hyperspectral imaging for disease detection, such as the need for accurate ground truth data and the influence of environmental factors on spectral signatures. However, with further research and advancements in technology, Hyperspectral imaging has the potential to revolutionize disease management in agriculture.

The article [35] proposes a deep learning-based approach for automated detection of yellow rust disease in winter wheat using high-resolution Hyperspectral UAV images. The model combines spectral and spatial information extracted from the images and utilizes multiple Inception-Resnet layers for feature extraction. The performance of the model is compared with a random forest classifier, and it is found that the proposed model achieves higher accuracy in detecting yellow rust disease. The study demonstrates the potential of the deep learning architecture for crop disease detection and suggests future work to validate the model on different datasets and develop efficient data analysis algorithms for large Hyperspectral images.

The article [36] discusses the use of deep learning algorithms for identifying wheat leaf diseases. The proposed algorithm, called RFE-CNN, combines a residual channel attention block, a feedback block, elliptic metric learning, and a convolutional neural network as shown in Figure 6. The algorithm extracts features from healthy and diseased wheat leaves, optimizes the features, and trains them using feedback blocks. The features are then processed and classified using a CNN and elliptic metric learning. The experimental results show that the proposed model outperforms other CNNs in terms of recognition precision, time consumption, and adaptive ability.

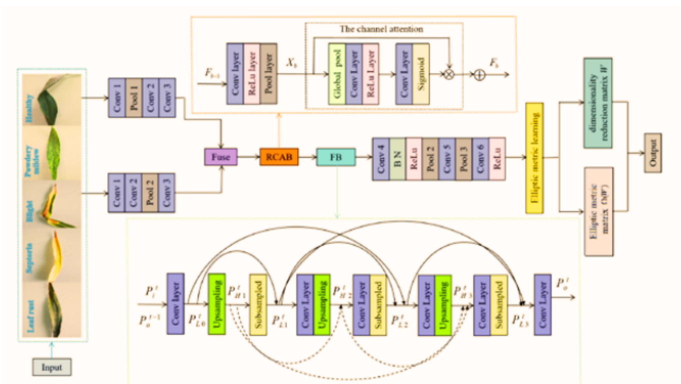

Fig. 6. Proposed overall classification framework[36].

The article [37] of the article is uses continuous wavelet analysis (CWA) to identify and discriminate between different stresses (powdery mildew, stripe rust, and nitrogen-water stress) in winter wheat at the canopy scale. The article also compares the performance of continuous wavelet features (WFs) with spectral bands (SBs) and vegetation indexes (VIs) in stress identification. Additionally, the article discusses the establishment of estimation models for disease index (DI) and validation.

The article [38] discusses a deep learning-based system for detecting wheat diseases. The authors propose a system that utilizes VGG19 deep convolutional neural network architecture and a dataset of photographs of healthy and diseased wheat plants. The pre-trained VGG19 model is refined and evaluated using various metrics such as accuracy, precision, recall, and F1 score. The results show promising performance in detecting and classifying wheat diseases, including powdery mildew, yellow rust, and leaf rust. We have mentioned more details about the article's results and methods in Table 01 and Table 02.

TABLE I
SOME RELATED WORKS ON HSI ON WHEAT CROPS

Reference	***Methods***	***Relevant Spectral Lines (nm)***	***Accuracy***
[23]	Analysis-Vegetation Indices-Separation	400-1000	Not Mentioned
[24]	Mask R-CNN, Gaussian Naïve Bayes, K-nearest neighbours, linear discriminant analysis, multi-layer perceptron neural network, random forests, support vector machine with linear kernel, support vector machine with radial basis function kernel, and gradient boosting	350-1050	97%
[25]	DCNN, DRNN with LSTM and GRU, and an improved hybrid CRNN, The 2D-CNN-BidGRU model	400-1000	F1 score:75%
[26]	Deep convolutional neural networks (CNNs)	spatial and spectral dimensions	(99.75%), purely spectral (86.5), purely spatial (98.70)
[27]	DarkNet 19, VGG16, and ResNet50	400-750	97%
[28]	a nearest-neighbour classifier50	360-900	96.9%
[29]	classification and regression tree (CRT)	400-1000	infected, wheat ears, 98.4%, and 80.8%, moderately infected leaves 87.8%
[30]	CFI system called FluorCam 700MF	550-560	96%
[31]	Principal Component Analysis (PCA), Successive Projections Algorithm (SPA), and Random Forest (RF), Support Vector Machine (SVM), and Naive Bayes (NB)	400-1000	96.44%
[32]	successive projections algorithm (SPA), (SVM), (NDVI), (MSR),	407-1000	95.8%
[33]	(SVM), random forest (RF)	500-620	96.75%
[34]	partial least-squares linear discrimination analysis (PLS-LDA)	400-1000	85%

TABLE II
ANOTHER RELATED WORKS ON HSI ON WHEAT CROPS

Reference	***Methods***	***Relevant Spectral Lines (nm)***	***Accuracy***
[35]	Deep convolutional neural network (DCNN)	450-950	85%
[36]	parallel convolutional neural networks (CNNs), VGG-19, GoogLeNet, Inception-V4, and Efficient-B7	Not mentioned	99.95%
[37]	Fisher's linear discrimination analysis (FLDA) and support vector machine (SVM)	615-696	78.1%, 95.6%, and 95.7%
[38]	VGG19 deep convolutional neural network architecture	Not mentioned	97.65%

TABLE III
ANOTHER RELATED WORKS ON HSI ON WHEAT CROPS

Ref	***Dataset***	***Year***	***No of Images***	***Type***
[45]	Wheat Yellow Rust Disease Infection type Classification	2021	268	Classification
[46]	Kaggle Wheat Leaf Dataset	2021	407	Classification
[47]	CGIAR Computer Vision for Crop Disease Dataset	2020	1486	Classification
[48]	Wheat Nitrogen Deficiency and Leaf Rust Image Dataset	2020	859	Classification
[49]	Crop Disease Treatment Dataset (CDTS)	2020	2353	Segmentation
[50]	NUST Wheat Rust Disease Dataset (NWRD)	2023	100	Segmentation

VII. DATASETS FOR HSI IN WHEAT DISEASES

Deep learning models are learned using varied datasets that are readily available and well-annotated. The development and testing of machine learning models for autonomous crop disease detection and diagnosis rely heavily on disease datasets. Open-source datasets on wheat disease are, however, hard to come by in the literature. The majority of the databases on wheat disease that are now accessible categorize wheat crops according to the disease condition, kind of disease, and disease severity level. The three most well-known large-scale plant disease datasets are PlantVillage [39], DeepWeeds [40], and PlantDoc [41], however, they are not specific to wheat crops. Images of wheat illnesses are present in only a small part of these datasets. There are several Hyperspectral imaging wheat disease datasets available for research purposes. One dataset includes hyperspectral images taken in a lab setting from wheat leaves and in situ from agricultural wheat fields and disease nurseries [42]. Another dataset contains hyperspectral images of four wheat lines, each with a control and a salt (NaCl) treatment [38]. Additionally, novel datasets provide valuable information for evaluating crop health and assessing the damage caused by diseases such as Fusarium head blight (FHB) in wheat kernels [43]. Another several datasets we presented in Table 3. Dataset characteristics are displayed in Table 3, including name, year of publication, number of images, and type (classification or segmentation). The number of leaves in each image increases the dataset's complexity

VIII. DISCUSSION

This study focuses on utilizing hyperspectral imaging and deep learning for identifying and mapping wheat diseases. It addresses challenges in wheat farming like pests, diseases, climate change, water scarcity, and yield improvement. The need for non-destructive, autonomous remote detection technologies with high sensitivity and reliability is underscored. The article explores the potential of hyperspectral imaging and deep learning for early disease detection, precise disease mapping, and objective disease severity assessment. The synergy of hyperspectral imaging and deep learning offers great promise in detecting wheat crop diseases. This technology enables high-resolution spectral data acquisition, aiding deep learning algorithms in disease identification and classification. A key advantage of using hyperspectral imaging and deep learning for wheat crop disease detection is its non-invasive nature. Unlike traditional methods involving manual inspection and sampling, this approach is less time-consuming and subjective. Additionally, hyperspectral imaging can capture detailed spectral information across various wavelengths. However, challenges exist, such as analyzing and interpreting large hyperspectral datasets and the necessity for robust training datasets. The effectiveness of deep learning models heavily relies on the quality and diversity of training data. In summary, combining hyperspectral imaging and deep learning shows promise for wheat crop disease detection.

IX. CONCLUSION

In this report, we highlight the importance of wheat as a major staple crop, as well as the difficulties associated with growing it due to pests and diseases. We stress the need for autonomous and non-destructive remote detection technologies that have high sensitivity and reliability to combat wheat infections. We identify hyperspectral imaging with deep learning as a powerful tool for early disease detection, precise disease mapping, and objective assessment of plant disease severity. The report provides a thorough explanation of algorithms, and data processing analysis, as well as articles on some prevalent illnesses of wheat that are detected using hyperspectral imaging we have provided some open datasets. We also discuss the benefits of hyperspectral technologies in accurately identifying and diagnosing various crop diseases. Lastly, we present the latest advances in early plant disease detection based on hyperspectral remote sensing. In conclusion, we find that hyperspectral imaging has great potential in disease detection in agriculture, especially in the field of crop diseases. It can help improve disease management strategies, optimize resource allocation, and minimize crop losses caused by plant diseases.

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